



PROJECT STATEMENT

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PROJECT: EXPENSE TRACKING SYSTEM

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Project Statement

Expense Tracker: A Menu-Driven Personal Finance Application in Python

1. Introduction

Managing personal money is a daily need, yet many people do not keep a record of what they spend. Without a record, it is hard to see where money goes, which makes saving and budgeting difficult. This project is a simple, easy-to-use expense tracking program written in Python that lets a user record, review, and analyse their spending from the terminal or any Python editor.

2. Problem Statement

Most people track expenses in their memory, on scraps of paper, or in complicated apps that need sign-ups, internet access, or a learning curve. These methods are unreliable, easy to forget, and often too heavy for someone who only wants a quick way to note down spending and see totals. There is a need for a **lightweight, offline, beginner-friendly tool** that:

- records expenses quickly,
- stores them safely between sessions,
- and shows clear totals and category-wise spending.

3. Objectives

1. To build a working expense tracker using only basic Python and its standard library.
2. To allow users to add expenses with an amount, category, note, and date.
3. To store data permanently in a CSV file so it is not lost when the program closes.
4. To display all expenses in a readable table along with the total spent.
5. To summarise spending by category so users can see where most of their money goes.
6. To let users delete incorrect entries.
7. To validate user input so the program does not crash on mistakes.

4. Scope

In scope

- A single-user, offline, command-line application
- Adding, viewing, summarising, and deleting expenses
- Persistent storage using a CSV file
- Input validation for amount and date

Out of scope

- User accounts or login
- Cloud sync or multi-device access

- Graphical or web interface
- Income tracking, budgets, or bank integration

5. Features

Feature	Description
Add expense	Records amount, category, optional note, and date (defaults to today)
View expenses	Shows all entries in a table with a running total
Category summary	Shows total per category with percentage of overall spending
Delete expense	Removes an entry by its number in the list
Data persistence	Saves automatically to expenses.csv
Input validation	Rejects non-numeric amounts, zero or negative amounts, and wrong date formats

6. Tools and Technologies

Item	Details
Programming language	Python 3.6 or newer
Modules used	csv, os, datetime (all built into Python)
Data storage	CSV file (expenses.csv)
Interface	Text-based menu in the terminal or editor console
External libraries	None required

7. System Design

7.1 Program Flow

Start

```

|
| Load expenses from expenses.csv
|
Show menu  <-----+
|
| User chooses an option
|
| +-- 1: Add expense -----> Save -----+
| +-- 2: View expenses -----+
| +-- 3: Summary by category -----+
| +-- 4: Delete expense -----> Save -----+
|                                     +-- 5: Exit

```

7.2 Data Structure

Each expense is stored as one record with four fields:

Field	Type	Example
Date	Text (YYYY-MM-DD)	2026-09-24
Amount	Number	250.0
Category	Text	food
Note	Text	Lunch

7.3 Main Functions

Function	Purpose
load_expenses()	Reads saved expenses from the CSV file
save_expenses()	Writes the current list back to the CSV file
add_expense()	Takes input, validates it, and adds a new record
view_expenses()	Prints all records and the total
show_summary()	Groups spending by category and prints percentages
delete_expense()	Removes a record chosen by the user
main()	Displays the menu and runs the program loop

8. Methodology

1. **Requirement analysis:** identify the basic needs of a personal expense tracker.
2. **Design:** decide on the menu structure, data fields, and storage format.
3. **Implementation:** write the program in Python using functions for each feature.
4. **Testing:** run every menu option with valid and invalid input, such as text instead of a number or a wrongly formatted date.
5. **Documentation:** prepare a README and this project statement.

9. Expected Outcome

A working Python program that any beginner can run without installing anything extra. The user can record daily spending in seconds, review past expenses, and understand their spending habits through the category summary.

10. Limitations

- Works for one user on one computer only
- Text-based interface, with no charts or graphics
- No editing of an existing entry (an entry must be deleted and added again)
- No monthly filtering or budget alerts
- Editing the CSV file by hand incorrectly can cause errors on load

11. Future Scope

- Monthly budget with warnings when the limit is exceeded
- Filtering by month or category
- Editing existing expenses
- Exporting reports
- Charts to show spending visually
- A graphical interface using Tkinter, or a web version using Flask

12. Conclusion

This project shows how a useful everyday tool can be built with only the basics of Python: functions, loops, file handling, and input validation. It is simple enough to understand fully, works reliably, and gives a solid base that can be extended with more features later.

THANK YOU!